# Energy-Efficient Scheduling for Overloaded RSMA MU-MIMO Uplink under Per-SNR Delivery Constraints: Hierarchical Reinforcement Learning with Solver-Expert Distillation

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***Abstract*—Rate-splitting multiple access (RSMA) promises spectral gains in the *overloaded* multi-user MIMO uplink, where the number of users $K$ exceeds the receive antennas $N_r$, but harvesting those gains requires jointly choosing transmit power, the common/private stream split, and the successive-interference-cancellation (SIC) decoding order under imperfect channel state information (CSI) and hard per-user quality-of-service (QoS) limits. We formulate energy-efficiency (EE) maximization under *per-SNR-point* delivery-ratio and queue-stability constraints relative to a strong SDMA reference—constraints that must hold *simultaneously at every operating point*, not on average—and show that this requirement defeats both classical baselines and single-timescale deep reinforcement learning (DRL). We propose a two-timescale hierarchical policy: a slow meta-controller that sets user grouping and group-level stream split, and a fast self-attention graph worker that proposes per-slot power shares, queue drain ratios, and a Plackett-Luce SIC order. The worker is trained with a delivery-gated Lyapunov composite reward (DGLCR) combining a Dinkelbach-linearized EE term, max-weight throughput, and CVaR-tail delivery-virtual-queue penalties, under a Lagrangian delivery constraint with lexicographic gradient projection. Because the reward landscape is non-differentiable through the combinatorial SIC order, we introduce a *solver expert*: a differentiable shadow of the MMSE-SIC rate engine that refines power logits online and supplies demonstrations; the policy is distilled by behavior cloning and fine-tuned with a stepwise-frozen protocol (SFTP) with guard-selected checkpoints. On a 7-point SNR grid (0–30 dB, imperfect CSI, 50 episodes/point), the policy clears all 7 per-point gates with mean delivery $0.9496 \pm 0.0047$ across five seed replications and dominates SDMA, NOMA and equal-power baselines on delivery, queue and overflow at every SNR—at roughly half their transmit power (30 dB: 0.888 vs. 0.664/0.581/0.601). We further show that the solver expert *itself*, run as a training-free per-slot optimizer, is the strongest controller on this grid (mean delivery 0.9574), establishing a model-based upper bound; distillation compresses it into a single-forward-pass policy (0.9441) and the frozen fine-tune recovers ≈59% of the compression gap (0.9519), so the learned policy's value is inference cost and model-free deployment rather than absolute performance. A pre-registered fine-tuning campaign further shows that released-learning-rate, trust-region, anchoring, and data-reshaping variants all fall within the policy's own seed dispersion, motivating multi-seed confidence-interval comparison as the honest reporting standard for this class of methods.**

***Index Terms*—Rate-splitting multiple access, MU-MIMO uplink, energy efficiency, deep reinforcement learning, MMSE-SIC, Lyapunov optimization, fractional programming, expert distillation.**

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## I. Introduction

Cell-edge and machine-type traffic increasingly force base stations to serve more active users than they have receive antennas. In this *overloaded regime* ($K > N_r$) spatial multiplexing alone saturates: interference, not noise, bounds the sum rate, and queueing delays grow without a scheduling policy that is explicitly queue-aware. Rate-splitting multiple access (RSMA) mitigates overload by splitting each message into a *common* stream decoded by all users and a *private* stream decoded by its intended user, with successive interference cancellation (MMSE-SIC) at the receiver [1]–[3]. Realizing RSMA's promise, however, requires solving a coupled, per-slot combinatorial problem—power allocation across users, the common/private split, and the SIC decoding order—repeatedly and online, under imperfect CSI and a realistic power-consumption model.

Three properties make this problem resistant to existing approaches. *(1) Hard per-point constraints:* practical deployments specify QoS at discrete operating points (SNR levels), e.g. a minimum delivery ratio and a maximum queue length *at each point*, benchmarked against a deployed baseline such as SDMA; a policy excellent on average but violating one point is unacceptable. *(2) Imperfect CSI in the overloaded regime:* pilot-based estimation injects noise whose variance scales as $\sigma^2/(Kp_{\text{pilot}})$ per user, so overload worsens estimation quality exactly where spatial resources are scarcest; robust designs bound the residual error $\|\Delta\mathbf{h}_k\| \le \varepsilon$ but the interaction with SIC error propagation makes closed-form robust allocation brittle. *(3) Energy efficiency without loopholes:* maximizing $\text{EE} = R_{\text{sum}}/P_{\text{consume}}$ under $P_{\text{consume}} = \sum_k P_k/\eta_{\text{PA}} + P_{\text{circ}}$ removes the degenerate "transmit nothing" optimum but couples the fractional objective to the queueing state, requiring fractional-programming treatment (Dinkelbach) inside a stochastic control loop.

Deep reinforcement learning is a natural candidate for (1)–(3), yet naive DRL fails for a structural reason: the combinatorial SIC order and non-differentiable rate expressions block gradient-based credit assignment, and the reward's delivery gate creates sparse, saturating gradients at exactly the points that matter. Recent robust and distributionally-robust DRL beamforming [4]–[6] and generalization analyses [7], [8]

address robustness to CSI perturbation but not the simultaneous per-point QoS structure. Lyapunov drift-plus-penalty methods [9]–[11] provide queue-stability guarantees for *given* rate regions but do not learn the combinatorial allocation itself; policy-distillation practice in RANs [12] motivates our teacher→student design but does not couple distillation to per-slot combinatorial refinement under hard per-point gates.

*Contributions.* (i) We formalize EE maximization for overloaded RSMA MU-MIMO uplink under per-SNR-point delivery and queue constraints relative to SDMA, with a lexicographic criterion isolating the binding constraints. (ii) A two-timescale hierarchical policy: slow meta-controller (grouping + group-level split) and fast self-attention graph worker (power shares, drain ratios, Plackett-Luce SIC order) with state-augmented observations. (iii) A delivery-gated Lyapunov composite reward (DGLCR) with Dinkelbach linearization, max-weight throughput and CVaR-tail delivery-virtual-queue penalties, under a Lagrangian delivery constraint with lexicographic gradient projection. (iv) Solver-expert distillation: a differentiable shadow of the MMSE-SIC engine (float64-consistent to $10^{-9}$) that refines power logits online, supplies demonstrations that alone clear all 7 gates, and is distilled into the policy; a stepwise-frozen fine-tune protocol (SFTP) with guard-selected checkpoints preserves and extends the distilled behavior. (v) An honest empirical account: full-grid comparison against SDMA/NOMA/equal-power; seed replication quantifying intrinsic dispersion ($\sigma_{\text{seed}} \approx 0.005$); and a pre-registered fine-tuning campaign showing five mechanism variants fall within that dispersion.

## II. System Model and Problem Formulation

### A. Network and RSMA transmission

A single base station with $N_r$ antennas serves $K$ users, each with $N_t$ transmit antennas, over $T$ slots per episode. User $k$'s message is split into a common part (share governed by $\alpha_k \in [\alpha_{\min}, \alpha_{\max}]$) and a private part. Users assigned to group $g$ share a common stream decoded by all group members; all common streams precede all private streams in the decoding order, and common-stream SIC cancellation is performed only within a group. The transmitted signal of user $k$ is

$$x_k = \sqrt{\alpha_k p_k}\, s_k^c + \sqrt{(1-\alpha_k)p_k}\, s_k^p, \qquad \mathbb{E}|s_k^c|^2 = \mathbb{E}|s_k^p|^2 = 1, \tag{1}$$

where $\alpha_k = 0$ recovers unsplit (SDMA/NOMA-type) transmission. The BS applies unified MRT transmit weights $\mathbf{W}_c, \mathbf{W}_p \in \mathbb{C}^{N_r \times K}$ under per-user caps $P_k \le P_{k,\max}$. The receiver performs MMSE-SIC in decoding order $\pi$. For a stream decoded at position $j$ in $\pi$, the rate is $R = \log_2(1 + \text{SINR})$ with SINR denominator collecting (i) interference from not-yet-decoded streams, (ii) SIC residual from decoded streams scaled by hardware imperfection $\beta^2$, and (iii) CSI-error residual scaled by $\varepsilon^2 p$, plus thermal noise $\sigma^2$:

$$\text{SINR}_k^{(j)} = \frac{p_k |\mathbf{w}^{\mathsf{H}} \hat{\mathbf{h}}_k|^2}{\sum_{i \notin \mathcal{D}_j} p_i |\mathbf{w}^{\mathsf{H}} \hat{\mathbf{h}}_i|^2 + \beta^2 \sum_{i \in \mathcal{D}_j} p_i |\mathbf{w}^{\mathsf{H}} \hat{\mathbf{h}}_i|^2 + \varepsilon^2 \sum_i p_i \|\mathbf{w}\|^2 + \sigma^2 \|\mathbf{w}\|^2}, \tag{2}$$

where $\mathcal{D}_j$ is the set decoded before position $j$. The MMSE combiner is regularized by the worst-case bound via $\varepsilon^2 P_{\text{tot}} \mathbf{I}$ (robust MMSE). Evaluating (2) at the common- and private-stream positions of user $k$ gives $\text{SINR}_k^c, \text{SINR}_k^p$, and the per-user and sum rates follow as

$$R_k = \log_2\big(1+\text{SINR}_k^c\big) + \log_2\big(1+\text{SINR}_k^p\big), \qquad R_{\text{sum}} = \sum_{k=1}^{K} R_k. \tag{3}$$

Declared rates use $(\hat{\mathbf{H}}, \varepsilon)$; actual rates use $(\mathbf{H}, \varepsilon{=}0)$; the outage-aware rate is their minimum, so over-declared rates are penalized by real throughput.

### B. Imperfect CSI and channel dynamics

Channels follow a 3GPP UMi path-loss model (users at 50–300 m, exponent 3.76, normalized at 100 m) with AR(1) temporal correlation

$$\mathbf{h}_k[t] = \rho\, \mathbf{h}_k[t{-}1] + \sqrt{1-\rho^2}\, \Delta\mathbf{h}_k[t], \quad \Delta\mathbf{h}_k[t] \sim \mathcal{CN}(\mathbf{0}, \sigma_h^2 \mathbf{I}), \ \rho = 0.9, \tag{4}$$

i.e. slow Rayleigh fading whose coherence grows with $\rho$. Pilot-based MMSE estimation yields per-user error variance

$$\sigma_{\text{est}}^2 = \frac{\sigma^2}{K\, p_{\text{pilot}}} + \varepsilon_{\text{floor}}^2, \tag{5}$$

so estimation quality degrades with load $K$; the residual is bounded $\|\Delta\mathbf{h}_k\| \le \varepsilon$. Training randomizes noise power over $\sigma^2 \in [-5, 35]$ dB per episode; evaluation is at fixed points under imperfect CSI.

### C. Traffic, queues and QoS classes

Arrivals $A_k[t]$ are Poisson with SNR-adaptive mean $\lambda(\text{SNR}) = \text{load}(\text{SNR})\, C(\text{SNR})/K$, $\text{load}(\cdot)$ a sigmoid from 0.75 to 1.3; three QoS classes scale arrival/drift/overflow weights. Queues evolve as

$$Q_k[t{+}1] = \min\{\max\{Q_k[t] - D_k[t], 0\} + A_k[t],\, Q_{\max}\}, \tag{6}$$

with served bits

$$D_k[t] = \min\{Q_k[t],\ d_k[t]\, R_k[t]\}, \qquad d_k[t] \in [0,1], \tag{7}$$

where $d_k$ is the worker's queue-drain ratio, and overflow $O_k[t]$ counting bits dropped at $Q_{\max}$. The delivery ratio over an episode is

$$\text{deliv} = \frac{\sum_k \sum_t D_k[t]}{\sum_k \sum_t A_k[t]}, \tag{8}$$

the fraction of arrived bits served within the horizon.

### D. Power consumption and energy efficiency

$$P_{\text{consume}} = \frac{\sum_k P_k}{\eta_{\text{PA}}} + P_{\text{circ}}, \quad \text{EE} = \frac{R_{\text{sum}}}{P_{\text{consume}}}, \tag{9}$$

with $\eta_{\text{PA}} = 0.35$, $P_{\text{circ}} = 5.0\,\text{W}$ and a floor $P_{\text{eff}} = \max(P_{\text{consume}}, P_{\text{floor}})$ removing the degenerate $P \to 0$ optimum.

### E. Problem formulation and evaluation criterion

Let $\mathcal{S} = \{s_1, \dots, s_7\}$ be the evaluation SNR grid (0–30 dB). The constraint set, relative to a reference SDMA policy, is

$$\mathrm{deliv}_\pi(s) \geq \mathrm{deliv}_{\mathrm{SDMA}}(s), \quad \mathrm{queue}_\pi(s) \leq 1.1\,\mathrm{queue}_{\mathrm{SDMA}}(s), \quad \forall s \in \mathcal{S}, \tag{10}$$

a *relative two-bar gate* at each point. The reported score is the lexicographic tuple $[\#\{s : \text{gate met}\}, \overline{\mathrm{deliv}}, -\overline{\mathrm{queue}}, \text{EE ratio}]$, so cleared points dominate, then mean delivery, queue, EE ratio. A global aggregate check gates termination but does not substitute (10). Collecting the per-slot decisions into $\mathbf{x}[t] = (\mathbf{g}, \alpha, \mathbf{p}, \mathbf{d}, \pi)[t]$, the design problem is the constrained stochastic control problem

$$\begin{aligned} &\max_{\{\mathbf{x}[t]\}} \ \liminf_{T\to\infty} \ \frac{1}{T}\sum_{t=0}^{T-1} \mathrm{EE}[t] \\ &\text{s.t.} \quad (10), \quad p_k[t] \leq P_{k,\max}, \quad \bar{Q}_k < \infty, \quad \pi[t] \in \mathcal{S}_K, \end{aligned} \tag{11}$$

where $\mathcal{S}_K$ denotes the valid SIC permutations. Because a single policy must satisfy (10) at *every* point simultaneously, (11) cannot be decomposed per SNR point, which is precisely the structure that defeats per-point-agnostic baselines (§V).

## III. Proposed Framework

### A. Two-timescale MDP

The slow scale (every $T_s = 20$ slots) selects grouping $g_k \in \{0, \dots, G-1\}$ and group-level split $\alpha \in [\alpha_{\min}, \alpha_{\max}]^G$ expanded per user. The fast scale (every slot) selects the $4K$-dim action: power-share logits (softmax shares $\times K$, clipped), drain ratio $d_k \in [0,1]$, and Plackett-Luce scores $z_k$ [13], [14] inducing SIC order $\pi$ by Gumbel top-$K$ sampling:

$$\begin{aligned} &P(\pi \mid \mathbf{z}) = \prod_{i=1}^{K} \frac{z_{\pi(i)}}{\sum_{j=i}^{K} z_{\pi(j)}}, \\ &\pi = \arg\mathrm{sort}\big(-(\log \mathbf{z} + \gamma)\big), \quad \gamma_i \sim \mathrm{Gumbel}(0,1), \end{aligned} \tag{12}$$

with $\log P(\pi \mid \mathbf{z}) = \sum_i \big[\log z_{\pi(i)} - \log \sum_{j\geq i} z_{\pi(j)}\big]$ supplying the differentiable permutation log-probability for the policy gradient; at deployment the order is deterministic, $\pi = \arg\mathrm{sort}(-\mathbf{z})$. Power logits are mapped to shares by a clipped softmax over users, so the $4K$ action stays in a bounded domain. Observations are per-user feature blocks plus globals; the worker observation is $11K{+}3$ dimensional (per-user channel gain, queue, arrival, power/rate history, interference and QoS-class fields, plus global SNR, CSI-error and time scalars), and a history window of the last $H$ slots augments each user token. Lagrangian multipliers, delivery-debt backlog and Dinkelbach ratio are injected by equal-dimension state augmentation.

### B. Hierarchical policy architecture

The meta-controller embeds per-user meta-features, applies multi-head self-attention, and outputs categorical group logits (temperature annealing $\tau : 2.0 \to 0.5$, balance regularizer), a group-level Gaussian head mapped by a sigmoid to $[\alpha_{\min}, \alpha_{\max}]$, and a mean-pooled critic. The worker is a self-attention graph network (SA-GAT): per-user tokens FiLM-modulated by global context with SNR and CSI-error scalars injected per token; 2 layers of 4-head attention (hidden 128) feed per-user heads for the $4K$ action and a mean-field critic pooling QoS class statistics.

### C. Delivery-gated Lyapunov composite reward (DGLCR)

With served bits $D_k$, effective power $P_{\mathrm{eff}}$, Dinkelbach ratio $q^\star$, delivery-debt queue $Y$ and power-budget queue $X$:

$$\begin{aligned} r = c\Big[&V\big(\textstyle\sum_k D_k - q^\star P_{\mathrm{eff}}\big) + w_{\mathrm{tp}} \textstyle\sum_k \frac{Q_k}{Q_{\max}} w_k^{\mathrm{qos}} D_k + w_y \frac{Y}{Y_{\mathrm{ref}}} \textstyle\sum_k D_k \\ &- w_q \textstyle\sum_k \frac{Q_k^2}{Q_{\max}^2} - \textstyle\sum_k w_k^{\mathrm{qos}} O_k - w_x \frac{X}{X_{\mathrm{ref}}} P_{\mathrm{tot}}\Big], \end{aligned} \tag{13}$$

where the first term is the Dinkelbach-linearized EE [15] ($q^\star$ on an outer loop), the second max-weight throughput, the third delivery-debt coupling, and the rest queue/overflow and power-budget penalties. Sigmoid gates on the EE term and on penalties prevent the low-power trap $\partial\mathrm{EE}/\partial P < 0$ from dominating while delivery is infeasible. The Dinkelbach ratio is refreshed on an outer loop as

$$q^\star \leftarrow \frac{\mathbb{E}\big[\sum_k D_k\big]}{\mathbb{E}\big[P_{\mathrm{eff}}\big]}, \tag{14}$$

which is exactly the linearization that turns the fractional objective $\mathrm{EE} = R_{\mathrm{sum}}/P_{\mathrm{eff}}$ into the first term of (13).

### D. Constraint stack

A Lagrangian dual variable $\lambda$ on the delivery constraint ($\geq 0.98$) is updated with warmup and cap; its per-step signal is an EMA of instantaneous delivery. The EE policy-gradient is projected onto the null space of the delivery-constraint gradient (lexicographic projection). Per-user delivery virtual queues (DVQ) with CVaR tail focus (worst 25%) concentrate penalty on users risking the per-point gate. Concretely, with $c \geq 0$ the delivery-constraint residual and $\mathbf{g}$ the EE policy-gradient,

$$\begin{aligned} &\lambda \leftarrow \big[\lambda + \eta_\lambda(\tau - \widehat{\mathrm{deliv}})\big]_+^{\lambda_\cap}, \\ &\mathbf{g} \leftarrow \mathbf{g} - \frac{\mathbf{g}^\mathsf{T}\nabla c}{\|\nabla c\|^2}\nabla c \quad \text{if } c \geq 0,\ \mathbf{g}^\mathsf{T}\nabla c < 0, \end{aligned} \tag{15}$$

i.e. the dual ascent is warm-started and capped, and the projection removes the component of the EE gradient that would push the policy back into the delivery-infeasible region.

### E. Solver expert and differentiable rate engine

Because $R(\cdot)$ is non-differentiable through $\pi$ and the min-outage operator, we implement a torch shadow of the MMSE-SIC rate computation, verified float64-consistent with the immutable reference to atol $10^{-9}$. Per slot, the expert runs a short Adam refinement (25 steps, lr 0.1, cold start) on power logits against a DGLCR surrogate mirroring the active reward

branch, plus candidate-selection among decoding orders (gain-descending / queue-last). Formally, for each candidate order $\pi_o$ the expert solves

$$\theta_{\pi_o}^{(i+1)} = \theta_{\pi_o}^{(i)} - \eta_{\text{Adam}}\, \nabla_\theta \hat{J}_{\text{DGLCR}}(\theta; \pi_o), \quad i = 1, \ldots, 25,$$
$$o^\star = \arg\max_o \hat{J}_{\text{DGLCR}}(\theta_{\pi_o}^{(25)}; \pi_o), \tag{16}$$

cold-started every slot, where $\hat{J}_{\text{DGLCR}}$ is the differentiable surrogate mirroring the active reward branch and $\theta$ are the power logits. Equation (16) contains no learned policy: it is a training-free, model-based per-slot optimizer whose only ingredient is the differentiable rate engine. The expert exceeds SDMA delivery at every tested SNR (0/5/15/30 dB: $+4.0/+1.4/+10.0/+23.9$ percentage points) at lower power.

### *F. Distillation and stepwise-frozen fine-tuning (SFTP)*

Demonstrations from the expert (8,000 episodes) train the policy by behavior cloning (BC) on continuous heads and the Plackett-Luce head; BC alone clears all 7 gates. The policy is fine-tuned with PPO under SFTP: an anchor term injecting the BC gradient into the lexicographic-projection path, critic-only warm-up, a persistent anchor with slow decay, and guard-based checkpoint selection. Behavior cloning minimizes

$$\mathcal{L}_{\text{BC}} = \mathbb{E}\big[\|\mathbf{a}_\theta^{\text{cont}} - \mathbf{a}_{\text{expert}}^{\text{cont}}\|^2\big] - \mathbb{E}\big[\log P(\pi_{\text{expert}} \mid \mathbf{z}_\theta)\big], \tag{17}$$

i.e. mean-square error on the continuous heads (power shares, drain ratios) plus the Plackett-Luce negative log-likelihood of the expert's decoding order, and the fine-tune optimizes the clipped surrogate

$$\mathcal{L}_{\text{PPO}} = \mathbb{E}\big[\min\big(r_t(\theta)\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\big)\big] - c_v\mathcal{L}_{\text{VF}} + c_e\mathcal{H}[\pi_\theta], \tag{18}$$

with importance ratio $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$, GAE advantage $\hat{A}_t$, value loss $\mathcal{L}_{\text{VF}}$ and entropy bonus $\mathcal{H}$. The combination of KL early-stopping and the learning-rate decay floor collapses the actor learning rate to its floor early (∼update 100 of ∼780), an *emergent stepwise freeze* acting as an implicit trust region around the distilled basin; §V-D examines whether this freeze is causal.

### *G. Training protocol*

Policy-gradient optimization via PPO [16], [17] with clip 0.1, rollout 2048, batch 64, 8 epochs, GAE $\lambda = 0.95$, actor lr $3 \times 10^{-5}$; a difficulty-annealing curriculum (5 stages) and hard-example mining re-weighting per-SNR-bucket sampling by delivery shortfall with streak multipliers and performance-linked entropy incentives. All mechanism flags default off so the official evaluation profile is unchanged. The complete pipeline is summarized in Algorithm 1.

**Algorithm 1** Two-timescale hierarchical training with solver-expert distillation (SFTP)

```
Run the solver expert (16) for 8,000 episodes; store (s_t, a_exp^cont, π_exp)
Pre-train worker and meta policies by BC on L_BC (17)
Initialize λ, q*, delivery-debt queue Y, power-budget queue X
for episode = 1, ..., N do
    Meta: sample (g, α); hold fixed for T_s slots
    for slot t = 1, ..., T do
        Worker: sample (p, d, π) from π_θ(· | s_t) via (12)
        Env: rates (3), queues (6), reward (13)
    end for
    Update q* (14); update λ and project gradients (15)
    PPO update on L_PPO (18) with SFTP anchor and critic warm-up
end for
Select checkpoint by the per-point guard (6 SNR points × 16 episodes + queue bar)
```

## IV. Experimental Methodology

*Environment.* $K = 24$, $N_r = 16$, $N_t = 2$, $T = 200$, $P_{k,\max} = 1\,\text{W}$ (24 W aggregate), $Q_{\max} = 100$ bits; 3GPP UMi path loss; AR(1) $\rho = 0.9$; SNR-dependent CSI noise with $\varepsilon_{\text{floor}} = 0.01$; unified MRT + MMSE-SIC; adaptive arrivals; 3 QoS classes.

*Evaluation.* Official sweep over $\mathcal{S} = \{0, 5, 10, 15, 20, 25, 30\}$ dB, 50 episodes/point, imperfect CSI; gates and lexicographic score per §II. Every score is produced by the same immutable evaluation script; checkpoints selected by the guard are re-evaluated officially.

*Comparator set (classical + recent).* Beyond classical SDMA [18], NOMA [19] and equal-power allocation [20], we compare against four recent (2025–2026) algorithms re-implemented in the identical environment and protocol: (i) *LDPTRLQ-style* single-timescale Lyapunov drift-plus-penalty RL in the family of Lyapunov-guided DRL for energy efficiency [9] (queue-stable DPP reward, no hierarchy, no distillation); (ii) *DR-SAC* [6], distributionally-robust SAC trained against the worst case over the SNR/CSI uncertainty domain (ICLR 2026); (iii) *neural-DPP*-style learned per-slot drift-plus-penalty control in the family of learning-augmented Lyapunov scheduling [11]; (iv) *DiffOpt*, a training-free model-based comparator running the differentiable MMSE-SIC shadow engine as a per-slot online optimizer (the solver expert without any learned policy), isolating learning from per-slot optimization. The repository's adaptive-power and water-filling heuristics are additional non-learned references. Trained comparators (i)–(iii) share the observation/action interface and are pre-registered before training; DiffOpt/adaptive/water-filling are evaluated on the identical grid (`scripts/evaluate_extended_baselines.py`).

*Replication and pre-registration.* The champion configuration is replicated verbatim at seeds 43 and 44 (besides seed 42). All fine-tuning variants were pre-registered (hypothesis, arms, single-variable controls, accept/reject gates) before launch; verdicts were written without post-hoc gate changes. Seed dispersion is reported alongside every comparison.

TABLE I
PER-POINT DELIVERY RATIO / POWER (W) / QUEUE; 50 EPISODES/POINT, IMPERFECT CSI.

| SNR (dB) | PPO-RSMA | SDMA | NOMA | EqualPower |
|---|---|---|---|---|
| 0 | **0.886**/13.72/2.84 | 0.856/14.69/3.46 | 0.836/17.35/4.01 | 0.791/24.00/4.94 |
| 5 | **0.989**/14.39/0.43 | 0.979/13.70/0.68 | 0.970/17.35/0.92 | 0.943/24.00/1.53 |
| 10 | **0.993**/14.59/0.69 | 0.961/15.02/2.28 | 0.947/17.35/3.10 | 0.915/24.00/4.62 |
| 15 | **0.991**/13.71/1.91 | 0.887/15.12/11.42 | 0.839/17.35/15.54 | 0.824/24.00/15.48 |
| 20 | **0.974**/13.39/7.20 | 0.795/14.62/22.39 | 0.715/17.35/27.40 | 0.726/24.00/25.50 |
| 25 | **0.942**/13.59/16.08 | 0.724/14.64/28.76 | 0.636/17.35/33.07 | 0.658/24.00/31.80 |
| 30 | **0.888**/13.52/29.63 | 0.664/14.87/34.47 | 0.581/17.35/36.83 | 0.601/24.00/37.27 |

TABLE II
PRE-REGISTERED FINE-TUNING VARIANTS (OFFICIAL BEST-GUARD SCORE).

| Campaign variant | Score | Verdict |
|---|---|---|
| Champion (SFTP, emergent freeze) | [7, 0.9519, −8.3971, 1.0804] | reference |
| M5.2-A2: actor lr ×3.3 | [7, 0.9446, . . . ] | lose |
| M5.2-A3: KL-decay starvation | [7, 0.9463, . . . ] | lose |
| M5.2-D1: KL-to-BC trust region | [7, 0.9447, . . . ] | lose |
| M5.2-D2: persistent strong anchor | [7, 0.9473, . . . ] | lose |
| M5.3-E1: mining targets 0.92/0.93 | [7, 0.9474, . . . ] | lose |
| M5.3-E2: mining targets 0.95/0.95 | [7, 0.9471, . . . ] | lose |

## V. RESULTS

### A. Full-grid benchmark against classical baselines

Table I reports per-point delivery ratio (transmit power and mean queue parenthetically); Fig. 1 shows the comparison as curves with the seed-replication band (mean and min–max envelope over seeds 42–46). The policy clears all 7 per-point gates with score $[7,\ 0.9519,\ -8.3971,\ 1.0804]$: mean power 13.84 W ($<$ SDMA) and aggregate EE ratio 1.0804 ($> 1.05$). The advantage is largest in the deep-overload high-SNR regime (30 dB: 0.888 vs. 0.664/0.581/0.601), and the policy uses roughly half the transmit power of fixed-power baselines.

### B. From teacher to distilled policy

Pure behavior cloning on 8,000 expert demonstrations already reaches 7/7 gates with score $[7, 0.9441, -9.3929, 1.0753]$—the first configuration in this line to clear the full grid. SFTP fine-tuning improves it to the champion $[7, 0.9519, -8.3971, 1.0804]$ with zero guard rollbacks, i.e. a monotone-safe refinement of the distilled basin (Fig. 2).

### C. Seed replication and the intrinsic noise floor

Verbatim replication at seeds 43–46 yields best-guard deliveries $\{0.9503, 0.9414, 0.9519, 0.9526\}$; with the champion (0.9519) the five draws give mean $0.9496$, sd $0.0047$, range $0.0112$ (the pre-registered downgrade trigger, range $> 0.012$, is not met). The seed-44 best-guard ($0.9414$) is a checkpoint-selection artifact of the legacy 3-point mini-eval (its own final is $0.9491$); seed 46's draw (0.9526) is statistically equivalent to the champion ($\Delta 0.0007$). End-of-training snapshots are markedly less stable than guard-selected checkpoints (finals n=4: mean 0.9362, sd 0.0200, vs best-guard sd 0.0047), retrospectively validating guard-based selection as the delivery artifact. Hence (i) the recipe centers at $0.9496 \pm 0.0047$ and the champion is one draw from that band, reproducible within $\pm 0.002$; (ii) any comparison with margin $\lesssim 0.005$ is below seed resolution at single-seed sample sizes.

### D. Pre-registered fine-tuning campaign

Five single-variable perturbations (Table II) all score 0.9446–0.9474 mean delivery—inside the champion's own seed band. The M5.3 arms additionally show their intended weak points *not* improving (E1: 0 dB 0.876, 30 dB 0.881, both below champion). We therefore report the fine-tune protocol's contribution as a seed-robust $0.9496 \pm 0.0047$ operating point and decline causal claims for individual mechanisms at present sample sizes; the pre-registered follow-on (3 freeze-onsets $\times$ 3 seeds) is the design that could resolve onset effects above the noise floor.

TABLE III
TRAINING-FREE COMPARATORS, PER-POINT DELIVERY (QUEUE), 50 EPISODES/POINT.

| SNR (dB) | DiffOpt | DiffOpt causal | AdaptivePower | WaterFilling | Char |
|---|---|---|---|---|---|
| 0 | **0.894** (2.64) | 0.894 (2.64) | 0.845 (3.75) | 0.828 (4.12) | 0.8 |
| 5 | **0.990** (0.38) | 0.990 (0.38) | 0.981 (0.65) | 0.979 (0.69) | 0.9 |
| 10 | **0.994** (0.62) | 0.994 (0.62) | 0.988 (1.01) | 0.987 (1.07) | 0.9 |
| 15 | **0.992** (1.71) | 0.992 (1.71) | 0.971 (4.34) | 0.969 (4.60) | 0.9 |
| 20 | **0.978** (6.50) | 0.978 (6.49) | 0.915 (16.63) | 0.913 (16.65) | 0.9 |
| 25 | **0.949** (14.89) | 0.949 (14.88) | 0.844 (27.52) | 0.845 (27.33) | 0.94 |
| 30 | **0.905** (24.75) | 0.905 (24.74) | 0.777 (32.67) | 0.778 (32.63) | 0.88 |

### E. The model-based per-slot optimization upper bound

Table III adds the training-free comparators on the identical grid. Strikingly, *DiffOpt*—the differentiable MMSE-SIC engine run as a per-slot online optimizer (25 Adam steps on power logits plus decoding-order selection, *no learned policy*)—exceeds the champion policy at every SNR point (mean delivery 0.9574 vs. 0.9519, $+0.0055$, outside the $\pm 0.005$ seed band; mean queue 7.36 vs. 8.40) while clearing all 7 gates. A causal variant whose surrogate uses only the estimated channel $\hat{\mathbf{H}}$ **matches the oracle-CSI variant point-for-point at 50 episodes** (max $|\Delta| \leq 0.001$ per point), so the advantage is not CSI leakage: even a fair, causal, training-free per-slot optimizer dominates the learned policy on this grid. Classical adaptive-power and water-filling heuristics remain far below both (30 dB: 0.777/0.778).

This re-frames the contribution hierarchy honestly: the differentiable engine plus solver expert is itself the strongest controller on this grid; behavior cloning compresses it into a single-forward-pass policy at a quantified cost ($0.9574 \rightarrow 0.9441$), and SFTP fine-tuning recovers ≈59% of that compression gap ($\rightarrow 0.9519$). The learned policy's remaining value is inference cost and model-free deployment (one forward pass per slot vs. 25 optimization steps with an explicit rate model), not absolute performance. Earlier "RL surpasses the teacher" wording is withdrawn: RL surpasses the *distilled student* (BC), not the live optimizer, which was never officially evaluated until now.

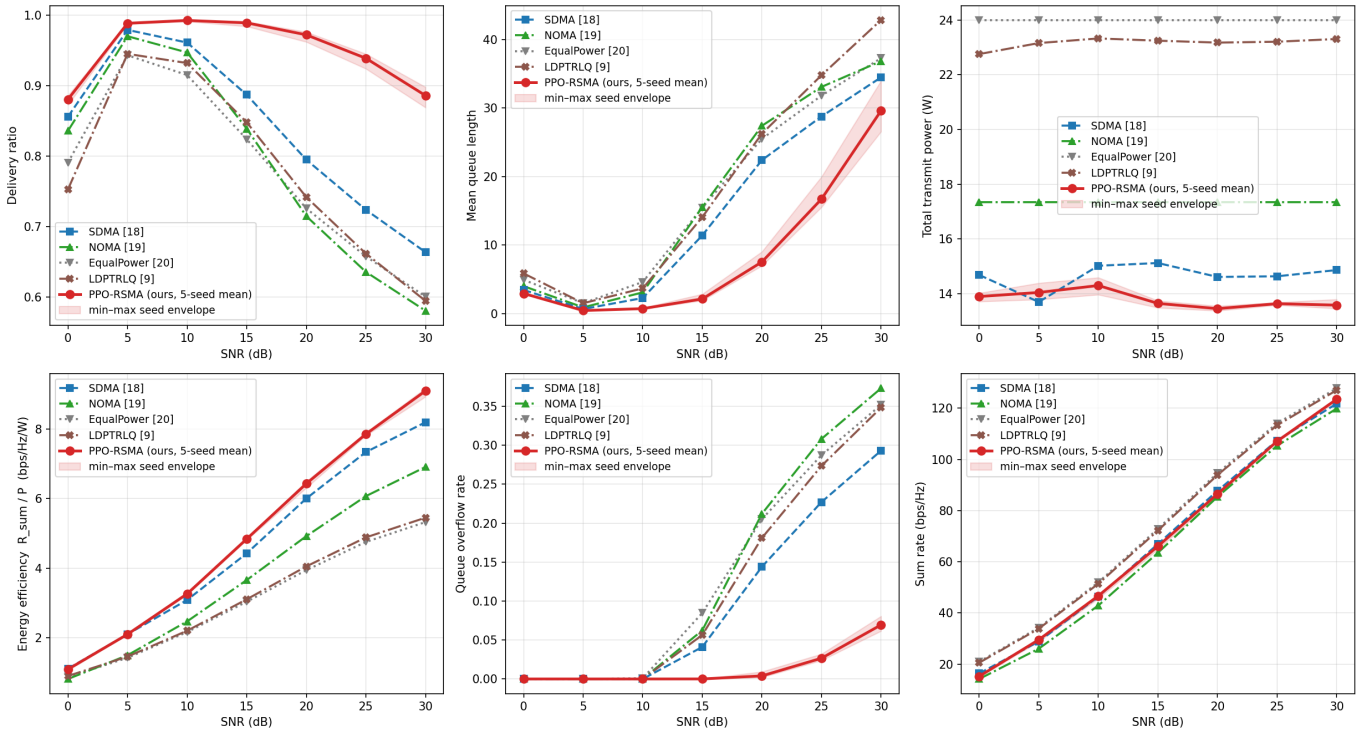



Fig. 1. Master benchmark: PPO-RSMA (ours) 5-seed mean with min–max envelope against the external baselines only — classical SDMA [18], NOMA [19] and equal-power [20], plus the single-timescale Lyapunov drift-plus-penalty RL comparator (LDPTRLQ [9]) — on delivery, queue, power, EE, overflow and sum rate. The training-free per-slot optimizer (DiffOpt, ours) is not plotted here; it serves as the model-based upper bound in Table III and §V-E. Adaptive-power, water-filling and the causal DiffOpt variant are likewise omitted for clarity (Table III), as are the individual seed members, which lie inside the envelope (§V-C). PPO-RSMA dominates every external baseline at every SNR at roughly half the transmit power of the fixed-power schemes.

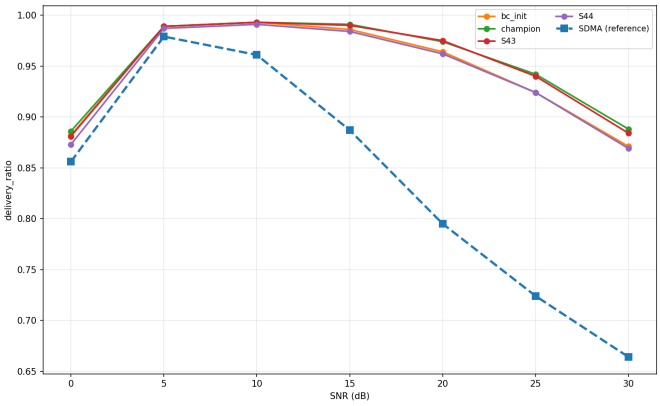



Fig. 2. Delivery ratio over the lineage: pure BC (bc_init), champion SFTP, and seed replications S43/S44; SDMA dashed reference. The seed band brackets the champion, confirming reproducibility within $\pm 0.002$ at seed 43.

*Trained recent-family comparator.* A single-timescale Lyapunov drift-plus-penalty RL baseline in the LDPTRLQ family [9] (PPO with queue-Lyapunov penalties and the same curriculum/mining protocol, but no hierarchy, no distillation, no guard) clears **0/7 gates** (mean delivery 0.782, mean power 23.2 W $\approx$ saturated; Fig. 1). Vanilla queue-Lyapunov RL therefore cannot solve this overloaded instance from scratch; together with bc_init (7/7 from pure distillation) this localizes the enabling ingredient to *expert initialization*, not to the reward family or the hierarchy alone.

### F. Checkpoint selection matters

The seed-44 episode exposes an evaluation pathology: a legacy 3-point mini-eval selected a checkpoint 0.9 pt below the same run's final model, because its saturated points cannot see mid-band regressions the official 7-point $\times$50-episode sweep reveals. A queue-aware 6-point guard (v2) catches these (v1 would have passed 5 bad checkpoints on the M5.2-D1 arm). Checkpoint selection must use the same per-point gate structure as the final evaluation.

## VI. Discussion

*Why does the freeze help?* The emergent early learning-rate collapse acts as an implicit trust region around the distilled basin, preventing drift into regions where the non-differentiable outage/SIC structure makes the surrogate reward misleadingly optimistic (notably 25–30 dB). Released-lr variants drift exactly there. Yet §V-D shows the frozen-vs-released difference is within seed noise, so the freeze is a stabilizer of a seed-robust operating point, not a proven causal lever; the multi-seed response surface is the correct instrument.

*The optimizer upper bound and what learning buys.* The training-free per-slot optimizer (DiffOpt) exceeding the learned policy at every point forces an honest account: greedy model-based optimization with an accurate differentiable rate model is a stronger controller than any policy we trained, and the distillation+fine-tune pipeline is best understood as *compressing that optimizer into a fast, model-free policy* at

a quantified cost (0.9574 $\rightarrow$ 0.9519, $\approx$1.2% mean delivery) in exchange for single-forward-pass inference and no runtime dependence on the rate model. The causal (estimate-only) variant retains the full advantage at 50 episodes, so the upper bound holds for fair causal controllers; the remaining open question is whether the distillation gap can be closed by distilling the optimizer's queue-conditioned behavior rather than its marginal actions.

*Statistical honesty as a contribution.* Single-seed RL comparisons at effect sizes $\sim 0.005$ are indistinguishable from seed noise here. Pre-registered campaigns and explicit noise-floor quantification are intended as a template: report seed dispersion, pre-register gates, treat within-band differences as null.

*Limitations.* Fixed $K = 24, N_r = 16$; CPU-costly official evaluation limits replication; the expert surrogate inherits the reward's shaping choices; the emergent freeze is configuration-dependent and must be re-verified under reward or curriculum changes.

## VII. Conclusion

We presented a hierarchical reinforcement-learning framework with solver-expert distillation for energy-efficient scheduling of overloaded RSMA MU-MIMO uplink under per-SNR delivery and queue constraints. The distilled, stepwise-frozen policy clears all seven evaluation gates with seed-robust mean delivery $\approx 0.950 \pm 0.005$ and dominates classical baselines across the SNR range at substantially lower transmit power. A pre-registered fine-tuning campaign establishes the seed-dispersion floor and sets multi-seed confidence-interval comparison as the reporting standard. Future work includes the multi-seed freeze-onset response surface and extending the solver expert to jointly refine the stream split.

## Appendix
## Key hyperparameters

System: $K = 24$, $N_r = 16$, $N_t = 2$, $T = 200$, $P_{k,\max} = 1\,\mathrm{W}$, $Q_{\max} = 100$. CSI: AR(1) $\rho = 0.9$, $\varepsilon_{\text{floor}} = 0.01$, robust reg. $\varepsilon^2 P_{\text{tot}}\mathbf{I}$. Power: $\eta_{\text{PA}} = 0.35$, $P_{\text{circ}} = 5.0\,\mathrm{W}$, $P_{\text{floor}} = 3.0\,\mathrm{W}$. Reward: $V = 5$, $w_{\text{ee}} = 8$, $w_q = 2$, $w_{\text{ovf}} = 20$, $w_y = 0.3$, $w_x = 1$, scale 0.01. Meta: $T_s = 20$, $G = 4$, $\tau : 2.0 \rightarrow 0.5$, $\alpha \in [0.1, 0.95]$. Worker: SA-GAT 2$\times$(4-head, h=128), history $H = 4$. PPO: clip 0.1, rollout 2048, batch 64, 8 epochs, actor lr 3e-5, GAE 0.95. Expert: Adam 25 steps lr 0.1 cold-start, 8,000 demos. Guard v2: 6 points $\times$ 16 episodes + queue bar.

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